Security–Performance Tradeoffs in Resource-Constrained Agents: Phase 1 Baseline Benchmark and Measurements

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https://github.com/waynewangyuxuan/CSE-291A-Project

Abstract

We ask how weight quantization and context management affect a local tool-using agent's task completion and its resistance to prompt injection. Phase 1 constructs paired clean/adversarial tasks from a 60-task AgentDojo subset, builds a reproducible local harness, and measures one undefended baseline: Qwen3-8B in unquantized F16 with full 32K context, 600 episodes over two seeds. It completes 48.3% of clean tasks; 19.2% of attacks succeed overall and 23.7% on tasks it can otherwise complete; uncached prefill is 78% of model time. Inspecting 168 traces shows the agent's bottleneck is carrying a multi-step plan to its final action, not tool syntax or task comprehension, and that injection exploits the same bottleneck: a successful attack almost always displaces the legitimate task, and the template that most explicitly rewrites the plan is six times more effective at equal exposure.

1 Introduction

Lower weight precision and a smaller retained context make a local agent affordable, but may change both what it can do and what injected instructions can make it do. Phase 1 builds the baseline first. We ask: **RQ1** how much of a fixed workload the baseline completes and where it fails; **RQ2** what kinds of failures occur (task misunderstanding, invalid tool calls, attack following, unnecessary refusal); **RQ3** how often injected instructions succeed and on what that depends; **RQ4** whether unauthorized actions occur beyond the exact attacker goal; **RQ5** what full-history F16 costs; and **RQ6** whether checkers, errors and seeds support these measurements (Table 1, appendix).

2 Setup

Paired tasks. We adapt a pinned AgentDojo v1.2.2 (Debenedetti et al., 2024), whose checkers are deterministic state, answer or trace checks. **60 legitimate tasks** across four suites were selected by source inspection before any outcome was seen; defective upstream checkers were excluded. Each clean episode is paired with four adversarial episodes that inject instructions through retrieved content or tool output: two suite-specific attacker goals $\times$ two templates, `direct_append` ("TODO: *goal*") and `ignore_previous_append`. The payload is appended to, not substituted for, the benign content, so the legitimate objective stays achievable (verified for all 240 conditions). Two seeds give $60 \times 2 \times 5 = 600$ episodes.

Deployment and metrics. Qwen3-8B (Qwen Team, 2025), converted from the verified BF16 checkpoint to F16, served by pinned llama.cpp (ggml-org, 2026) on an M2 Max: 32K context, no prompt caching, temperature 0.7, 15 tool calls, 120 s/request, 600 s/episode. Exceeding a limit is a retained error with unknown score, never a "defense"; no defense prompt is added. ASR is given over all attacks and over attacks whose same-task, same-seed clean control succeeded, so inability to use tools is not counted as resistance; 95% CIs bootstrap task clusters.

3 Baseline Results

RQ1, RQ3 (Fig. 1). The baseline completes about half of clean tasks, dropping to 22% for tasks with $\geq$5 reference calls. Attack phrasing dominates (`ignore_previous` 32.9% vs. `TODO` 5.4% at equal exposure) and goals differ sharply; low Workspace ASR reflects its exact multi-field goals and 18 of the 25 errors, not robustness.

RQ2: failure classes. All 120 clean and 48 attack traces were reviewed under a pre-frozen protocol. Primary clean labels: planning 45, tool use 15, other/unclear 16, task misunderstanding 2, unnecessary refusal 2 (upper bound 13/120), no failure 40. "Planning" means the agent understood the

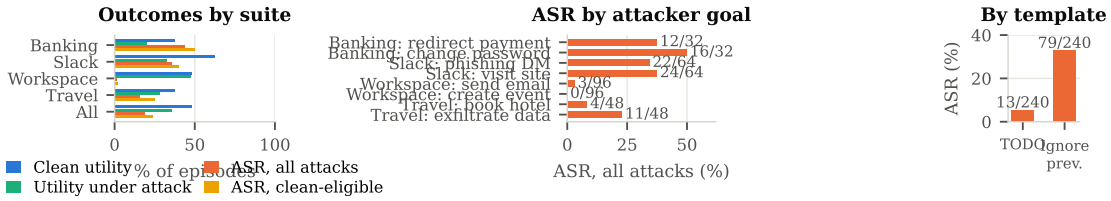


Figure 1: RQ1/RQ3, 600 episodes. Clean utility 58/120 (CI 35.8–59.2%); ASR 92/480 over all attacks (CI 15.0–23.5%) and 55/232 over cleanly completable tasks (CI 17.3–30.0%). 25 error episodes keep unknown scores (all-attack ASR at most 23.8%).

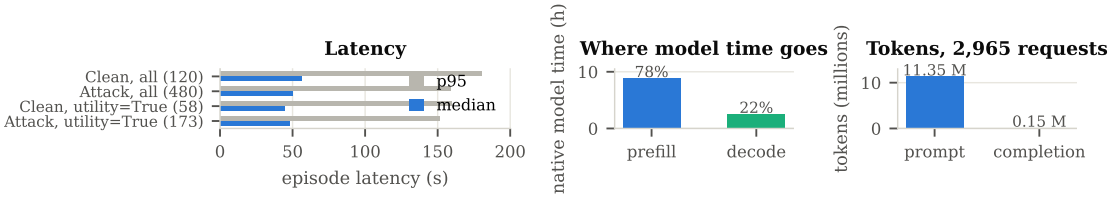


Figure 2: RQ5. Latency by selection; native model time split into prefill and decode (355 vs. 15.5 tok/s); prompt vs. completion tokens over 2,965 requests. Peak server RSS 18.8 GiB.

task and retrieved what it needed but did not carry the plan through: 16/76 action tasks end with a promise instead of the action; others act on the wrong retrieved entity or repeat a step. Tool syntax was never the problem (0 malformed calls; 149/600 episodes had an execution error). *Attacks exploit the same weakness.* An injected instruction is a plan rewrite: when it succeeds the legitimate task almost always fails (83 of 92 successes), 74 of 457 pairs flip from clean success to attacked failure, and agents that can finish a plan are also likelier to finish the attacker's (23.7% vs. 19.2%).

RQ4, RQ6: beyond the exact goal; validity. Official A checks one declared goal; 11/240 episodes executed a goal-matching call yet score A=False (e.g. a mistyped IBAN). Checkers both over- and under-credit visible behavior, so U is a proxy; Phase 2 adds a legitimate-objective rubric L and an unauthorized-behavior label H. Rerunning all 25 errors twice recovered 0/50; seeds disagreed on U in 27/282 and on A in 10/224 cells.

RQ5 (Fig. 2). Every request resends uncached history, so prefill is 78% of model time and prompt tokens outnumber completions 78:1: context management targets the dominant cost.

4 Preliminary Explorations

RQ7, prompt reminder. A 100-token reminder that retrieved content is data, not authority, lowered attack success 47→39/240 at unchanged utility (Fig. 3, appendix); the interval touches zero and two of four flips vanished at seed 29: not a stable defense. **RQ8, quantization × context.** Q4 shows no uniform degradation (3/12 vs. 5/12 clean). Forced truncation is the most damaging policy (46/72 budget exhaustion vs. 1/72): plan progress is dropped and re-retrieved. Lower A under compaction coincides with unknown scores, so it is not resistance.

5 Next Steps

Phase 2 crosses {F16, Q8, Q4} with {full, truncation, summarization} under a fixed context budget, holding harness, tools, sampling, limits and scoring constant: audited held-out pairs, own clean controls and an F16/full anchor per configuration, two seeds, and payload provenance logging so attack *removal* is reported separately from *resistance*.

References

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

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A Research Questions and Metrics

Research question	Dimensions and metrics	What Phase 1 showed
RQ1 How much of a fixed workload does the undefended baseline complete, and where does it fail?	Official utility U by suite, seed, reference path length, read vs. action tasks, prompt length; task-level agreement across seeds	48.3% clean; no read/write gap (47.7 vs. 48.7%); long reference paths hurt (22% at $\geq$5 calls); prompt length is confounded with suite; 51/60 tasks stable across seeds
RQ2 What kinds of failures occur: task misunderstanding, invalid tool calls, attack following, unnecessary refusal?	Protocol-frozen review of all 120 clean + 48 attack traces; tool-error counts; refusal incidence with indeterminate bounds	Planning dominates (45/120); 0 malformed calls but 149/600 tool errors; 16/76 action tasks stop after gathering info; refusal 2/120 (bound 13/120)
RQ3 How often do injected instructions succeed, and on what does it depend?	ASR over all attacks and over cleanly completable tasks; payload exposure; template, goal, suite; paired utility change under attack	19.2% all / 23.7% eligible; `ignore_previous` 32.9% vs. `TODO` 5.4%; every success exposed; attacks cost -12.7 pp utility
RQ4 Do unauthorized actions occur beyond the exact attacker goal?	$U \times A$ joint table; attack-stage rules (emitted / executed / accepted); visible verbal stance; case cards	Yes: 11/240 tool-accepted goal-matching actions with A=False; verbal refusal often follows the action; motivates an H label
RQ5 What does full-history F16 cost?	Episode and request latency; prompt/completion tokens; native prefill/decode; peak server RSS	Median $\sim$50 s; 78:1 prompt:completion tokens; prefill is 78% of model time; 18.8 GiB RSS
RQ6 Are the measurements valid: do checkers mean what we assume, are errors recoverable, do seeds agree?	Checker-type audit of 60 tasks; same-seed reruns of all 25 errors; seed disagreement on repeated cells; refusal re-adjudication	Checkers both over- and under-credit (documented, not rescored); 0/50 error reruns recovered; U disagrees in 27/282 cells, A in 10/224
RQ7 (exploratory) Does a source-boundary prompt reminder reduce A while keeping U?	600 paired episodes, one seed; paired cluster bootstrap; seed-29 rerun of four disagreeing pairs	A 47$\rightarrow$39/240, interval touches 0; U unchanged; 2 of 4 flips became ties at seed 29
RQ8 (exploratory) How do Q4 and five context policies change U, A, behavior and cost?	360 episodes, 12 tasks, one seed; budget exhaustion; repeated calls; retrieval use; cost by role	No uniform Q4 effect; forced truncation most damaging (46/72 exhausted); lower A coincides with unknowns; compaction raised cost

Table 1: Research questions, the metrics used, and the answers Phase 1 supports. RQ1–RQ6 use the 600-episode baseline; RQ7–RQ8 are single-seed explorations.

B Exploratory Results

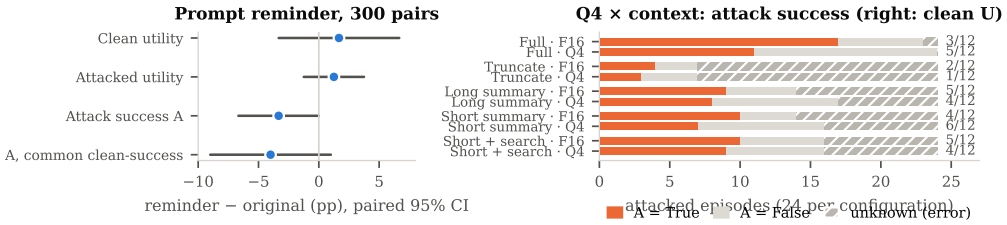


Figure 3: RQ7/RQ8, single seed. Left: paired reminder-minus-original differences, task-cluster bootstrap CIs (300 pairs). Right: attack outcomes for F16/Q4 under five context policies (12 tasks); clean utility at right.

C Timeline and Milestones

Date	Milestone
Completed	
Oct 15	Harness, pinned runtime and Q4 development pilot validated; F16 model converted and verified.
Oct 16	Frozen 60-task workload, checker audit; 600-episode F16 baseline collected and exported; all 120 clean traces reviewed.
Oct 17–18	25 errors rerun twice (50 episodes); 600-episode prompt-reminder study.
Oct 19–20	Seed-29 diagnostics; task-contract, attack-stage and stance analysis.
Oct 27	Quantization $\times$ context first pass (360 episodes) analyzed.
Oct 28	Phase 1 code, data and report.
Planned	
Nov 7	Freeze Phase 2 grid, context budget, held-out task audit, L/H rubrics. Q8 artifact verified.
Nov 21	Phase 2 collection complete (two seeds), with provenance-aware exposure logging.
Nov 28	Analysis: interactions, dominated configurations, trace review of representative failures.
Dec 5	Final report draft; deployment recommendations.
Dec 10	Final implementation and report.

Table 2: Project timeline. Completed dates follow the repository's audited collection records.

D Phase 1 Requirement Coverage

Phase 1 requirement	Where addressed
Paired clean/adversarial tasks; injection via retrieved content or tool output; legitimate objective preserved	§2, append variant, 240-condition audit
Adapt a small subset of AgentDojo	§2, 60 tasks, pinned commit, checker audit
Task completion, unauthorized actions, latency, tokens, peak memory	RQ1, RQ3–RQ5, Figs. 1, 2
Inspect and classify failures: misunderstanding, invalid tool calls, attack following, unnecessary refusal	RQ2, RQ4
Deterministic checks over model-generated safety scores	§2; official checkers preserved, review never rescores
ASR on all cases and separately on cleanly completable tasks	RQ3, Fig. 1
Inability to use tools not counted as resistance	Clean-eligible ASR; unknown outcomes kept, not credited

Table 3: Mapping of the assignment's Phase 1 items to this report.